USN. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[22BSPH11]**

SECOND SEMESTER B. Tech EXAMINATION, JUNE - 2023

ENGINEERING PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: I. Missing data may be suitably assumed.

II. Physical constants

- **i.** Mass of the electron $m_e = 9.1 \times 10^{-31} kg$
- **ii.** Mass of proton, neutron $m_n, m_p = 1.67 \times 10^{-27} kg$
- **iii.** Charge of the electron $e = 1.602 \times 10^{-19} C$
- **iv.** Planck's constant $h = 6.626 \times 10^{-34} Js$
- **v.** Velocity of light $c = 3 \times 10^8 m/s$
- **vi.** Boltzmann's constant $k = 1.38 \times 10^{-23} J/K$
- **vii.** Acceleration due to gravity $g = 9.8\, m/s^2$

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	State Heisenberg's uncertainty principle.	L-2	CO1	**02**
	b.	Define stimulated emission of radiation.	L-2	CO2	**02**
	c.	What are extrinsic semiconductors? Give an example.	L-1	CO3	**02**
	d.	List out the different types of Mach regimes based on Mach number	L-2	CO4	**02**
	e.	Discuss the different types of beams.	L-2	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	What are matter waves? Outline the characteristic properties of the matter waves.	L-1	CO1	**05**
		OR			
	b.	The ground state energy of an electron in an infinite well is 5.6MeV. Calculate the energy in eV of first excited state. What will be the ground state energy if the width of the well is doubled?	L-3	CO1	**05**
3.	**a.**	What is population inversion? Describe how the population inversion can be achieved using the metastable state.	L-2	CO2	**05**
		OR			
	b.	Discuss the different types of optical fibers with neat diagram.	L-3	CO2	**05**
4.	**a.**	Discuss the aspects in which the classical free electron theory has failed.	L-2	CO3	**05**
		OR			
	b.	Outline the assumptions of quantum free electron theory.	L-2	CO3	**05**
5.	**a.**	Explain the construction of working of light emitting diode.	L-2	CO4	**05**
		OR			
	b.	Differentiate between intrinsic and an extrinsic semiconductor.	L-2	CO4	**05**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus

6.	a.	What is simple harmonic motion. Derive the differential equation for simple harmonic motion.	L-2	CO5	**05**
		OR			
	b.	What are shock waves? Explain the properties of Shock waves.	L-2	CO5	**05**
7.	a.	Define the following. **i.** Stress **ii.** Strain **iii.** Young's modulus **iv.** Rigidity modulus **v.** Bulk modulus	L-1	CO6	**05**
		OR			
	b.	State and explain Hooke's law.	L-2	CO6	**05**
		PART-C **6 X 10=60**			
8.	a.	**i.** Describe Davison and Germer's experiment to show that the electron behaves as a wave.	L-2	CO1	**07**
		ii. Calculate the de-Broglie wavelength of the de-Broglie wave of an electron having a kinetic energy of 100 eV.	L-3	CO1	**03**
		OR			
	b.	Using one dimensional time independent Schrodinger's wave equation, obtain the wave function for a particle in a potential box of width "a" and hence normalize the wave function.	L-3	CO1	**10**
9.	a.	**i.** Describe the construction and working of CO2 laser with necessary diagrams.	L-2	CO2	**07**
		ii. Find the number of photons emitted per second by a He-Ne laser source emitting light of wavelength 632.8 nm with an optical power of 10mW.	L-3	CO2	**03**
		OR			
	b.	**i.** Deduce an expression for Numerical aperture in terms of refractive indices of core and cladding of an optical fiber and hence give the condition for propagation.	L-3	CO2	**06**
		ii. Calculate the numerical aperture, relative refractive index difference, V-number and the number of modes in an optical fiber of core diameter 50μm, core and cladding refractive indices as 1.41 and 1.40 at the wavelength of 820 nm.	L-3	CO2	**04**

10.	a.	i. Explain the variation of Fermi factor with temperature and energy.	L-2	CO3	06
		ii. Find the temperature at which there is 1% probability that a state with an energy 0.5eV above Fermi energy is occupied.	L-3	CO3	04
		OR			
	b.	i. Explain how the quantum free electron theory successfully overcome the failures of classical free electron theory.	L-2	CO3	06
		ii. Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material.	L-3	CO3	04
11.	a.	Discuss the procedure to find the band gap of a semiconductor experimentally with necessary equations.	L-2	CO4	10
		OR			
	b.	i. Using the expression for charge career concentration for electrons and holes, show that Fermi level lies exactly in between the valence band and conduction band in case of intrinsic semiconductor.	L-2	CO4	06
		ii. Use the following given data for intrinsic germanium at 330 K to find the resistivity. $n_i = 2.4 \times 10^{19}/m^3$, $\mu_e = 0.39\ m^2v^{-1}s^{-1}$, $\mu_h = 0.19\ m^2v^{-1}s^{-1}$.	L-3	CO4	04
12.	a.	Obtain an expression for force constant for springs when i. springs are connected to each other in series combination ii. springs are connected to each other in parallel combination	L-2	CO4	10
		OR			
	b.	Deduce an expression for amplitude and phase of forced vibrations.	L-2	CO5	10
13.	a.	Obtain the relation between Young's modulus, rigidity modulus and Poisson's ratio when the force is applied on one face of a cube.	L-3	CO6	10
		OR			
	b.	i. Obtain an equation for the Young's modulus when a downward force is applied to beam of a single cantilever. (Use the relation for the bending moment $= \frac{Y}{R} I_g$)	L-3	CO6	07
		ii. When a tangential force of 100 N is applied on one of the faces of cube, it displaces by 2 mm. If the side of a cube is 10 cm, calculate the rigidity modulus of a cube.	L-3	CO6	03